Central Venous Catheters Insertion and Management Procedure

1. Scope

Site	Operational Area	Applicable to
Armadale Kalamunda Group	All services	Nursing and Medical

2. Purpose

To establish minimum practice standards for the insertion, management and removal of Central Venous Catheters (CVC) throughout Armadale Kalamunda Group (AKG).

This procedure excludes:

- Implanted venous or arterial access ports.
- Peripherally Inserted Central Catheters (PICC).
- Catheters tunnelled under the skin on the chest wall and inserted into the subclavian vein.

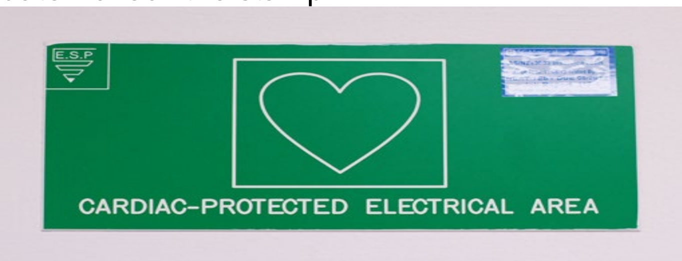
All health care professionals are to work within their scope of practice appropriate to their level of training and responsibility.

Electrical Safety

Most patient treatment rooms are 'body-protected electrical areas'. This means that precautions have been taken to prevent accidental macro shocks (greater than 10 mA) (Figure 3). However, a small shock well below this threshold, transmitted directly to a patient's heart via a guidewire could easily trigger a fatal arrhythmia. For this reason, any central line insertion should be conducted in a 'cardiac-protected electrical area' where precautions have been taken to prevent even micro shocks (greater than 100 μA). Look for the appropriate electrical safety sign.

Within AKG, CVCs are only to be inserted in ED, ICU or Theatre.

Cardiac-Protected Sign
Area suitable for CVC insertion provided that the electrical safety test is within date - check the stamp.



Body-Protected Sign
Area not suited for CVC insertions.
Suitable for most other patient treatments

Specific Considerations:

- Inserting, managing and removing patients' vascular access devices (VAD's) are only to be undertaken by persons with the appropriate training in the procedure, or under the direct supervision of an appropriately trained person.
- All Central Venous Catheters are to be reviewed daily and removed when no longer clinically indicated.

3. Procedure

3.1 Associated infection prevention equipment considerations

- When applying CVC catheter access hub antiseptic solution ensure to:
  - Determine compatibility with the indwelling CVC, e.g. aqueous or alcohol based solution.
  - Utilise a residual antiseptic where indicated for specific CVC.
- For insertion site hair removal use surgical clippers.
  - Do not shave the area with a razor.
- Aseptic non-touch technique and / or maximum barrier precautions.
- Situational requirements for utilising personal protective equipment (PPE).
- CVC inserted in emergency situations without appropriate infection prevention strategies in place at the time are to be removed within 24 hours.
  - A new CVC may then be inserted under appropriate conditions.

3.2 Equipment management considerations

- 5 Lumen Central Lines are preferred
- Select the PPE that are appropriate.
- Consider patient latex, dressing, surgical adhesive tape or other sensitivities.
- Selecting safety sharps to minimise the risk of sharps injury.
- Consider local anaesthetic where appropriate to the type that is least invasive for the specific device insertion.
  - **Do not use** local anaesthetic combined with adrenaline.
- All local anaesthetics used in device insertion must be appropriately prescribed.
- Use syringes 10 mL or larger, one dedicated per lumen for access.
- Intravenous sodium chloride 0.9% (5-30 mL) is the preferred diluent and may be administered without prescription to maintain CVC patency.
  - Consider compatible diluents when flushing CVC prior to and post prescribed medication administration.
- Choose a sterile, chlorohexidine gluconate impregnated gel pad self-adhesive dressing (Tegaderm CHG) for covering the insertion site as first preference or use a chlorhexidine gluconate impregnated foam disk (BioPatch) with a sterile transparent, semipermeable self-adhesive dressing.
- Ensure that the blue side of the BioPatch disk is facing up and white foam side down and the split in the disc lines up with catheter and makes full unbroken contact with skin at the insertion site.

  - Use sterile gauze where the patient is diaphoretic, or there is blood, or ooze from the insertion site.

3.3 Equipment

- CVC transducer set, if central venous pressure monitoring is requested.
- CVC Draping or Basic ICU pack.
- Skin antiseptic solution (minimum 2% chlorhexidine in 70% alcohol).
- Waterproof sheet (bluey).
- Sterile gauze.
- Ultrasound machine + sterile probe cover and gel.
- Adhesive tape.
- Sutureless securement device (or suture material and suture scissors if required).
- Local anaesthetic.
- 1x 3mL and 5mL syringes and 1x 21g and 25g needles if not in pre-prepared pack.
- 2x 10mL sodium chloride 0.9% syringes and 2x 10mL syringes.
- Sterile dressing (Tegaderm CHG) or BioPatch and Tegaderm.
- Sharps disposal container.
- Monitoring equipment.
- Oxygen therapy / sedation if required.
- Proceduralist PPE: surgical cap, mask and protective eye wear, sterile gown, towel and gloves.
- Assistant PPE: surgical cap and mask.

3.4 Staffing requirements

- A minimum of one staff member skilled to undertake the procedure and one for assistance.
- Assistance is required to maintain maximum aseptic and barrier precautions.
- Only appropriately trained medical staff should insert CVC's or they should be providing direct supervision.

4. Insertion procedure

- Check consent and the post insertion x-ray request form.
- Check the coagulation profile and full blood count.
- Determine the insertion site and catheter type.
- Ensure appropriate patient monitoring equipment available:
  - Obtain baseline observations.
  - Perform and document patient observations during procedure.
- Position patient appropriately:
  - Supine head down position, if tolerated and clinically appropriate.
  - Supine (without hip flexion) for femoral approach.
- Provide oxygen therapy where clinically indicated.
- Assemble the required equipment:

    - Setup using aseptic non-touch technique.
    - Check required medications, local anaesthetic and sedative.
    - Setup ultrasound equipment.
- Proceduralist to surgically scrub.
    - Ensure to don gown, gloves, eye wear, cap and mask.
- Swab site with skin antiseptic solution.
    - Minimum 2% chlorhexidine in 70% alcohol.
    - Allow to air dry.
- Drape the area for insertion.
- Administer local anaesthetic to site.
- Insert catheter using the Seldinger technique.
- Secure the CVC in position.
- Attach needleless adapters or Positive Pressure Valve (PPV) to all the lumen.
- Dress insertion site using a sterile dressing (Tegaderm CHG) or BioPatch and Tegaderm.
- Dispose of equipment appropriately, remove gloves and perform hand hygiene.

4.1 Key points post insertion

- Following any procedure where the catheter remains indwelling, ensure the device is well secured and dressing intact.
- The ultrasound probe must be decontaminated in accordance with the Decontamination of Diagnostic Ultrasound Transducers Procedure
- Specific CVC Insertion Monitoring include:
    - Heart rate, blood pressure and respiratory rate.
    - Cardiac monitoring, as clinically indicated.
    - Oxygen saturation.
    - Pain and discomfort.
- Arrange for post-insertion radiology, and follow-up result to confirm tip placement and identify a pneumothorax or haemothorax.

4.2 Potential problems during or post procedure

During Procedure	Post Procedure: Catheter related
Arterial injury	Breakages
Cardiac arrhythmias	Infection
Haemorrhage	Malposition
Haemothorax	Migration
Nerve damage	Occlusion
Pneumothorax	Thrombosis

5. Documentation

- Perform and document patient observations as clinically indicated and to ensure safe post procedure progress. If physiological observations are abnormal, refer to escalation process
- For all CVC procedures ensure to document:

  - Date, time and details specific to the procedure undertaken, including:
  - Actual type, size and distinctive features of the device inserted or accessed.
- Details of aseptic technique, e.g. emergency insertions.
- Insertion site identifiers, i.e. anatomical location, and/or marked drawings.
- The line depth must be recorded on the Central Venous Catheter form (AKMR121.7)
- Any attempted or incomplete procedures, e.g. unsuccessful insertions.
- Outcome of insertion site assessment.
- Details of the device patency and condition.
- Rationale regarding the need for device commencement, continuation or removal.
- Complete Central Venous Catheter form (AKMR121.7) and include within the patient health record.
- Document the external length of the catheter (skin to hub)

6. Dressings

6.1 Daily review of the CVC

- Assess the area surrounding CVC insertion site at each shift and during dressing changes for signs of
  - Localised infection: tenderness, pain, redness and swelling.
  - Catheter position: signs of migration. Document the external length each shift.
  - Securement and dressing integrity.
- The catheter insertion site should be visually inspected or palpated through the intact dressing to determine tenderness during daily review.
- Assess patient for signs of systemic infection: tachycardia, pyrexia, and hypotension.
- Assess the patency of lumens.
- Patients with femoral inserted CVC are not to flex hip.
- Determine ongoing need for CVC access.
- Inform medical officer or senior nurse if signs and symptoms of infection are present and determine management plan.
- Provide patient education to report any changes to insertion site or discomfort.
- Document details of daily review.

6.2 Dressing Change

- Frequency is determined by dressing type and condition:
  - Every seven days or earlier if damp, loose or soiled for the chlorohexidine impregnated dressing as well as for the dressing with chlorohexidine gluconate impregnated foam disk (BioPatch).
  - Every two days for sterile gauze dressing.
- Immediately if the dressing integrity is compromised, moisture, blood or infection present.
- Suture less securement device; if insitu, requires changing every seven days.
- Cover the CVC dressing with waterproof cover when showering.

7. Flushing and changing positive pressure valve procedure

Change positive pressure valves every 7 days

7.1 Indications

- To assess and maintain CVC patency prior to infusions.
- Following infusions to prevent mixing of incompatible medications or solution.

7.2 Procedure

- Perform hand hygiene and assemble the equipment.
    - Draw up syringes.
    - Prime the PPV.
    - Perform hand hygiene
    - Working one lumen at a time.
    - Clean existing PPV and lumen with antiseptic swabs.
        - Allow to air dry.
    - Remove and discard the existing PPV and replace swiftly with new primed PPV.
    - Insert flush syringe into the PPV, flush the lumen using a pulsatile method (push-pause) until lumen is clear.
        - Repeat for each lumen.
        - Discard equipment.
        - Perform hand hygiene

8. Blood sampling

8.1 Indications

- For patients where no other alternative exists.
- Following placement of a new device.

8.2 Blood sampling advice

- Utilise the largest lumen available.
- If multiple lumen catheter insitu, dedicate one lumen to purpose.
- Undertake blood sampling from lumen with no continuous infusion in progress, if possible.
- Flush line prior to blood sampling.

8.3 Blood culture sampling

- To determine suspected central-line blood stream infection, undertake:
    - One sample collected from the existing device, and
    - One sample collected from the separate peripheral venepuncture.

8.4 Blood Sampling Equipment

- PPE: Non-sterile gloves.
- Specimen request form.
- Required vacuum blood specimen tubes.

- 21g needle.
- 10 mL Luer lock syringes and 20 mL Luer lock syringes.
- Sharps disposal container.
- Kidney dish.
- Positive pressure valve.
- Antiseptic solution swabs / alcohol swabs.
- Sodium chloride 0.9% ampoules for injection:
  - Minimum 20 mL for flushing.
- One of the following type of blood collection device:
  - 1 x Vacutainer holder + Vacutainer Direct Draw Adapter – Assembled.
  - 1 x syringe (minimum 10 mL) + 1 x Vacutainer Blood Transfer Device.

8.5 Blood Sampling Procedure

- Confirm correct patient details with the pathology request form.
- Perform hand hygiene, clean work surface and assemble equipment.
- Cease infusion if in progress and if clinically appropriate.
- Perform hand hygiene and don gloves.
- Clean the PPV with antiseptic swabs.
  - **Allow to air dry.**
- Flush lumen with 10 mL sodium chloride 0.9%, attach 10 mL syringe into PPV and withdraw 10 ml blood from the CVC.
- Discard the syringe containing blood.
- Clean the PPV with antiseptic swabs.
  - **Allow to air dry.**
- Then withdraw the required amount of blood.
- 10 mL (minimum) syringe and 20 mL syringe for blood culture.
- Attach Vacutainer to syringe, then insert required specimen tubes.
- Tubes must be collected in the correct order as per the PathWest Recommended Order of Draw, Appendix C: Combined Tube Guide / Order of Draw for Ward Staff PSCP036
- Remove the syringe / blood device from PPV and place samples in kidney dish.
- Flush the lumen with 10-20 mL sodium chloride 0.9% using a pulsatile method (push-pause) until lumen and the PPV are clear.
- Clean the PPV with antiseptic swabs.
  - **Allow to air dry.**
- For blood collected via syringe for Blood cultures:
  - Clean the top of blood culture bottles with alcohol swabs.
  - Attach 21 g needle to syringe, place 10 mL of blood into bottle.
  - Repeat with new 21g needle for the second bottle.
- Ensure the CVC is secure, remove and discard the existing PPV and replace swiftly with a new primed PPV.
- Discard the equipment.

- Perform hand hygiene

9. Removal

9.1 Considerations

- Removal of the CVC requires precautions to prevent air embolism during and following procedure.
- Medical staff are to review full blood count and coagulation profile before procedure.
- When localised and / or systemic infection suspected, liaise with medical officer and consider obtaining insertion site MCS and device tip culture and / or blood cultures.

9.2 Equipment required

- Dressing pack.
- Alcohol swabs.
- Small occlusive dressing.
- PPE: Goggles / visor.
- Sterile and non-sterile gloves.
- Skin antiseptic solution.
- Dressing solvent if available.
- Sharps disposal container.
- If Infection Suspected:
    - Pathology request form.
    - Sterile specimen jar.
    - Sterile scissors.

9.3 Removal procedure

- Perform hand hygiene
- Prepare equipment.
- Position patient in supine, as tolerated.
- Perform hand hygiene
    - Don PPE.
- Using dressing solvent if available, remove and discard dressing:
    - Remove securement device using alcohol swabs and discard.
- Do not leave the patient.
- Perform hand hygiene (apply alcohol gel and allow to air dry).
- Don gloves (use sterile gloves if tip culture required).
- Create sterile field using aseptic non-touch technique between device and operator.
- Clean device insertion site.
    - Allow to air dry.
- Place gauze over device insertion site in preparation for removal.
- When removing catheters, it is also recommended to raise CVP by keeping the patient in **a supine position**.

- Instruct the patient to perform a Valsalva manoeuvre during catheter removal, if possible. If this is not possible, **removing the catheter during active expiration is recommended**.
- Removal should be timed to occur **during expiration** for patients who are not **on a ventilator** or at **end inspiration if on a ventilator**. This is when intrathoracic pressure is greater than atmospheric pressure to reduce the risk of aspirating air into venous circulation.
- If catheter difficult to remove, stop the procedure and consult a medical officer.
- Ensure that the **exit site is covered with impermeable dressing** and that pressure is applied afterward for 5–10 min, for haemostasis and prevention of bubble entry. It is recommended that the patient remains supine for 30 min after central venous access removal.
- Apply a sterile occlusive dressing and remain intact for a minimum of 24 hours.
- Prior to repositioning the patient following removal, ensure the dressing is airtight and occlusive.
- Observe device integrity.
- If device not intact consult the medical officer for further management and undertake physiological observations.

9.4 Post procedure

- For blood cultures, label the sample taken from the CVC.
- Complete a pathology request from and send samples to laboratory.
- Document details of procedure.

10. Legislative context

The following documents are required to give effect to this procedure:

- Health Department of Western Australia (HDWA) Consent to Treatment Policy MP 0175/22
- HDWA Clinical Handover MP 0095
- HDWA Recognising and Responding to Acute Deterioration Policy MP 0171/22
- East Metropolitan Health Service (EMHS) Patient Identification and Procedure Matching EMHS: 21

11. Supporting information

The following documents support the implementation of this procedure:

- Central Venous Catheter Form (AKMR121.7)
- Hand Hygiene Policy AKG-POL-0187-17
- Aseptic Technique Procedure AKG-PRO-0045-14
- Specimen Collection and Pathology Results Procedure AKG-PRO-0421-13
- Recognition and Response to Acute Deterioration Policy AKG-POL-0388-14

12. Compliance, monitoring and evaluation

Compliance will be evaluated through audit and monitoring of blood stream infections via Healthcare Infection Surveillance Western Australia (HISWA). Routine monitoring of related clinical incident monitoring via the DATIX-CIMS system.

13. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 2 – Action 2.5 Healthcare rights and informed consent
- Standard 3 – Action 3.8 Hand hygiene
- Standard 3 – Action 3.9 Aseptic technique
- Standard 6 – Action 6.5 Correct identification and procedure matching
- Standard 6 – Action 6.7 Clinical Handover
- Standard 8 – Action 8.4 Recognising acute deterioration

14. References

1. PathWest Recommended Order of Draw, Appendix A: Combined Tube Guide / Order of Draw for Ward Staff
2. Ingram P., Sinclair L., & Edwards T. (2006). The safe removal of central venous catheters. *Nursing Standard. 20(49),* 42–46. https://www.proquest.com/openview/38911c03e5e40405451166c100fbe413/1?pq-origsite=gscholar&cbl=30130
3. NSW Agency for Clinical Innovation. (2021). *Central Venous access devices (CVAD): Clinical practice guide.* https://www.aci.health.nsw.gov.au/__data/assets/pdf_file/0010/239626/ACI14_CVAD-2-2.pdf
4. Le fevre, P. L. (2021). *Central Venous Catheter Insertion guide.* Philippelefevre.com. Retrieved 6 May, 2021, from https://philippelefevre.com/downloads/cvc_insertion_guide.pdf

15. Glossary

The following definitions are relevant to this policy.

Term	Definition
Central Venous Catheter (CVC)	A CVC is a single or multi-lumen catheter inserted percutaneously (non-tunnelled) in the arm or chest into a large vein to enable venous access as appropriate for: • Intravenous fluid and medication administration. • Blood product administration. • Nutritional support. • Emergency vascular access.

16. Policy owner (relating to these procedures)

Enquiries relating to this procedure may be directed to:

Title: Nurse Unit Manager
Department: Intensive Care Unit
Email:

17. Review

This procedure will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V3.0	19/09/2024	19/09/2027	Scheduled review (*refer to policy bulletin or amendment form for specific details*)

18. Authorisation

This procedure has been authorised and issued by the A/Director Nursing and Midwifery.

Authorised by	Denver Prince – A/Director Nursing and Midwifery
Authorisation date	04/06/2024
Published date	19/09/2024

Appendix 1: Indications / contraindications for CVC procedure

Type	Description	Indications	Contraindications

CVC: Central Venous Catheter	Catheter inserted percutaneously via the subclavian, internal or external jugular, or femoral veins extending approx 15-30 cm with the tip positioned within the superior or inferior vena cava.
	One - three weeks short term access.
		• Where PIVC or PICC is impractical.
		• Frequent blood sampling.
		• Haemodynamic monitoring.
		• Administration of:
		  • Total parenteral nutrition (TPN).
		  • Non-isotonic and vesicant solutions (e.g. inotropes or chemotherapy).
		• Repeated IV fluids, blood products or medications.
			Severe bleeding diathesis

Complications troubleshooting guide

Complications	Signs and symptoms	During insertion	Post insertion

Pneumothorax	• Restlessness or anxiety
	• Dyspnoea
	• Cyanosis
	• Pain on breathing
	• Oxygen desaturation
	• Wheeze
	• Decreased air entry
	• Increase peak Airway pressure if ventilated
	• Decreased tidal volume
	• Asymmetrical chest wall movement
	• Surgical emphysema
		• Monitor patient's respiratory rate, oxygen saturation and general condition.
		• If concerned, discontinue procedure.
		• Obtain urgent CXR.
			• Monitor patient's respiratory rate and general condition.
			• Routine post insertion CXR if subclavian or jugular vein insertion.
			• **If signs or symptoms evident**:
			  • Sit the patient upright.
			  • Administer high flow oxygen.
			  • Seek urgent medical review.

Complications	Signs and symptoms	During insertion	Post insertion
Air embolism / catheter disconnect	• Restlessness or anxiety • Dyspnoea / cyanosis • Hypertension / dizziness • Tachycardia / weak pulse • Altered consciousness (e.g. confusion) • Visible catheter damage or leakage from line during use.	During insertion: • Position patient supine (with head down 30-degree tilt where possible). • Ensure all CVC connections are secure.	• Check all CVC connections and prevent further air entry (cap line with new positive pressure valve). • Position patient in left Trendelenburg (head down at 30-degree tilt) • Administer high flow oxygen. • Seek urgent medical review. • Monitor vital signs. • Complete Clinical Incident Form. **If visible damage:** • Clamp CVC between patient and damaged area, cover with sterile gauze. • Minimise patient movement. • Seek medical review and expert advice on management. • **Note**: Removal or repair of tunneled catheters should only be undertaken by specialist staff.
Arterial puncture	• Bright red blood • Syringe fills more quickly than expected (when compared to venous) • Blood leaves in a pulsing mode	• Apply direct pressure to site for five minutes to limit haematoma formation. • Monitor vital signs • Document in health record • Inform medical staff if tachycardia or hypotension occurs.	N/A
Cardiac tamponade	• Chest tightness or pain • Shortness of breath • Muffled heart sounds • Altered consciousness (as tamponade impairs myocardial contractility and	• Administer high flow oxygen • Monitor vital signs • Documented in health record • Activate MET or Cardiac arrest response if appropriate.	• Monitor clinical conditions and vital signs as indicated. • Request MO review as indicated by patient's condition.

Complications	Signs and symptoms	During insertion	Post insertion
	result in loss of cardiac output)	• Prepare for pericardiocentesis (as directed by MO)	
Hematoma formation	• Local swelling • Patient complains of pressure	• Record and monitor symptoms • Seek medical help if condition advances	• Same as for during insertion. • Vigilance is required in patients with coagulopathy
Catheter dislodgement	• External portion of catheter is visibly longer • Sudden lack of blood return • Swelling in arm, neck or chest	N/A	• Do not attempt to push catheter back into place or use it. • Seek guidance from MO.
Catheter occlusion	• Inability to flush catheter • Blood in lumen • Inability to aspirate blood • **Note**: partial obstruction may allow infusion of fluid but not aspiration • Cause of internal occlusion: • CVC migration to smaller vein or resting on vein wall • Compression of CVC in superior vena cava between the clavicle and the first rib • Constant wearing may result in damage to the integrity of the catheter • Other causes include chemical precipitation, lipid disposition or thrombus formation		• Routine use of push/pause technique for lumen flushes **External occlusion**: • Check for kinks (or clamp) and reassure line if indicated **Internal occlusion** • Ask patient to cough, take deep breaths, hunch shoulders or reposition • Attempt **gentle** irrigation with sodium chloride 0.9% using push / pause method alternating flushing and aspirating; do not exert excessive pressure or suction • If unable to remedy, or patient has signs or symptoms of a thrombus seek urgent MO review • CXR may be required to confirm tip position/exclude migration • If thrombus suspected – see below

Complications	Signs and symptoms	During insertion	Post insertion
Thrombus	• Swelling • Pain, numbness or tingling • Coolness, swelling or venous engorgement, discoloration of neck, chest or arm (on side of CVC)		• Patients at high risk of CVC thrombus (e.g. patients with cancer) maybe prescribed routine anticoagulation • Seek urgent MO review • Monitor vital signs • Consider need for oxygen therapy (if appropriate) • Consider fibrinolytic agent (e.g. Alteplase) as prescribed by MO
Infection	Site infection and tunnel infection: • Local pain or inflammation • Cellulitis or tracking within 2cm of exit site • Exudate or pus at site • Fever Systemic infection: • Fever (usually >38°C • Malaise • Rigors (particularly on flushing line) • Chills • Hypotension • Other signs of shock may develop so close observation is required • Elevated WCC, Raised CRP	Compliance with CVC Bundles • Maximal barrier precautions on insertion (use of mask, cap, sterile gloves and large drape) • Hand hygiene • Skin anti-sepsis with Chlorhexidine 2% in 70% alcohol • Optimal selection insertion site, avoid femoral approach • Use chlorhexidine bio-patch to reduce the regrowth of microbes after skin antisepsis	• Review need for CVC daily • Other routine care including hand hygiene, site monitoring, line change / lumen flushing as detailed If local infection suspected: • Inform MO and Shift Coordinator, review management plan • Review need for CVC • Take swab of site if exudate present • Consider antibiotic therapy • Document in health record If signs of systemic infection: • Inform the MO and Shift Coordinator and review management plan • Monitor closely for signs of shock • Remove CVC if no longer needed • Obtain paired blood cultures (from CVC and peripheral vein) prior to commencing antibiotics